

MA378: Tutorial Sheet 4

.....

Q1. Deduce the 4-point Newton-Cotes Rule for estimating the integral $\int_0^1 f(x)dx$:

$$Q_3(f) = q_0 f(x_0) + q_1 f(x_1) + q_2 f(x_2) + q_3 f(x_3).$$

Extend the rule to estimate the integral of functions over $[a, b]$.

Q2. Prove the error bound given for the Trapezium rule. That is, show that

$$\left| \int_a^b f(x)dx - Q_1(f) \right| := \mathcal{E}_1 \leq \frac{(b-a)^3}{12} M_2.$$

Q3. Explain clearly, with an example, why in general it is not true that $Q_n(f) \rightarrow \int_a^b f(x)dx$ as $n \rightarrow \infty$.

Q4. (i) Deduce an error estimate for the Composite Trapezium Rule.

(ii) Taking $N = 10$, give an upper bound for the error in the Composite Trapezium Rule when approximating $\int_1^2 \ln(x)dx$.

(iii) What value of N would you have to take to ensure that the error was less than 10^{-5} ?

Q5. (i) Deduce the formula for the *composite Simpson's Rule*.

(ii) Derive an error estimate for the *composite Simpson's Rule*.

(iii) What value of N would you have to take to ensure that the error in the estimate of $\int_1^2 \ln(x)dx$ is less than 10^{-6} ?

(iv) Denote the $(N+1)$ -point Composite Simpson's Rule by $S_N(f) \approx \int_a^b f(x)dx$. Show that, for sufficiently smooth $f(x)$,

$$\lim_{n \rightarrow \infty} S_N(f) = \int_a^b f(x)dx.$$

Q6. Determine the precision of the following schemes for estimating $\int_0^1 f(x)dx$.

(i) $Q(f) = f(\frac{1}{2})$. [Note: closely related to a question on Assignment 2, so won't be done in tutorials]

(ii) $Q(f) = \frac{1}{4}f(0) + \frac{3}{4}f(\frac{2}{3})$.

(iii) $Q(f) = \frac{3}{2}f(\frac{1}{3}) - 2f(\frac{1}{2}) + \frac{3}{2}f(\frac{2}{3})$.

Q7. (Based on the 2023/24 exam) Consider the following rule for approximating $\int_0^4 f(x)dx$:

$$R(f) = q_0 f(0) + q_1 f(1) + \frac{2}{9}f(3) + f(4).$$

(a) Determine the values of q_0 and q_1 so that $R(\cdot)$ has precision 1.

(b) What is the maximum precision of $R(\cdot)$ for the values of q_0 and q_1 that you have determined?

(c) Why is $R(\cdot)$ not a Newton-Cotes rule, in the strictest sense?

Q8. Use a change of variables, as we did with the Trapezium rule, to show that the rule for approximating $\int_0^1 f(x)dx$ is

$$G_1(f) = \frac{1}{2} \left(f\left(\frac{1}{2} - \frac{1}{2\sqrt{3}}\right) + f\left(\frac{1}{2} + \frac{1}{2\sqrt{3}}\right) \right).$$

More generally, extend the rule to an arbitrary interval $[a, b]$.

Q9. Use $G_1(x)$ to estimate $\int_1^2 \ln(x)dx$. How does this compare with the Trapezium and Simpson's Rule?

Q10. Derive a 3-point Gaussian Quadrature Rule to estimate $\int_{-1}^1 f(x)dx$. *Hint:* $x_1 = 0$.